



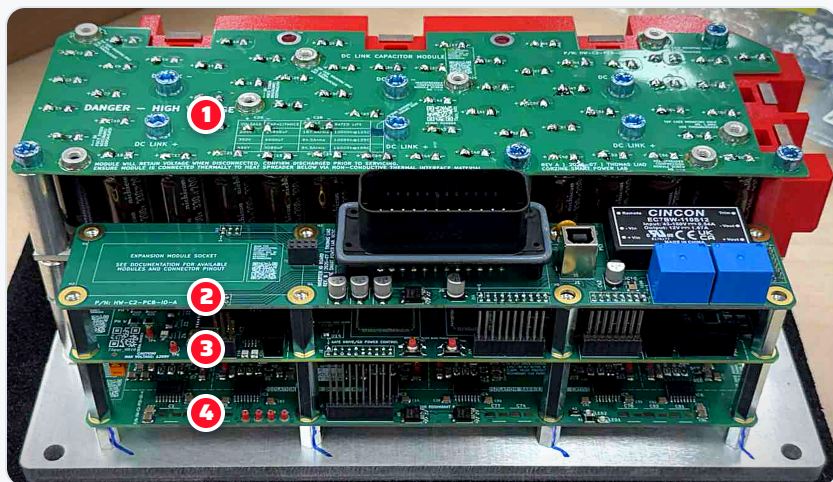
OpenVVF

The open-source, safety-engineered power conversion platform — 50 kW to 1 MW+.

Power electronics underpin electrification — yet above hobby scale, almost none of it is open. **Sealed firmware and proprietary hardware lock out the researchers, educators, and small manufacturers** who would otherwise build, study, and improve on it. OpenVVF is the open alternative: one reusable control module plus swappable power stages — motor drive to bidirectional DC-DC, IGBT today, SiC-ready.

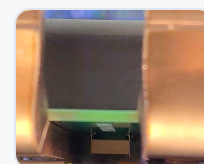
THE PLATFORM

One control module, a stack of replaceable boards



- 1 DC-link capacitor board**
Electrolytic cans (heatsunk to the baseplate); film + ceramic caps beneath — MLCCs soldered directly between the IGBT terminals for an ultra-short commutation loop.
- 2 IO board**
USB-B, dual CAN FD, and the expansion socket for application IO.
- 3 Control board — dual-MCU**
Main MCU runs FOC motor control; an independent safety coprocessor monitors safety-critical outputs and can bring the system to a safe state.
- 4 Gate-drive board**
Isolated gate drive and sensing for the three IGBT half-bridges.

Below the stack, **three IGBT half-bridge modules conduction-cool to an aluminum baseplate** — no liquid loop on Size 2. The same control module runs every chassis: **motor self-commissioning** (resistance / inductance calibration for PMSM and induction machines), the **RTE Studio** model-based workflow — fully open-source, no MATLAB required, with a software-in-the-loop (SIL) pipeline in progress — and the browser-based **Telemetry Viewer**. The controls program has also benchmarked **ADRC and MPCC against the production FOC**.



Ceramic MLCCs between IGBT terminals.

PLATFORM FAMILY

One control module — four power stages

CONTROL MODULE Dual-MCU control module

FOC motor control + independent safety coprocessor · USB-B / dual CAN FD / expansion · RTE Studio + Telemetry Viewer

Size 1

~50 kW class inverter
Compact chassis for light EVs and lab benches.

● EARLY DESIGN

Size 2 Inverter

~100 kW · 400 V / 465 A
Built and bench-tested; validation under way.

● IN VALIDATION

Size 2 DC-DC

Charging, storage & microgrids
Same inverter blocks, liquid-cooled enclosure.

● IN PROGRESS

Size 3

MW class · 900 V / 1400 A
All-film DC-link; arc-flash-analyzed enclosure.

● DESIGN IN PROGRESS

Built, bench-tested, on the record

400 V class

465 A continuous · 600 A peak (60 s)
· 150/200/400 V by DC-link selection

10

Formal test reports · every one links
its interactive telemetry log

PMSM + induction

Motor self-commissioning,
instrument-verified

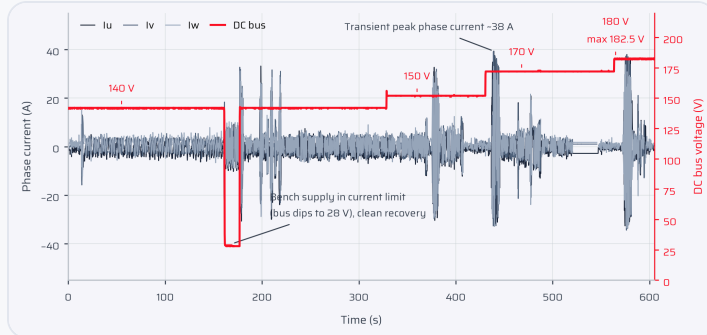


Fig. 1 — 180 V power-stage bring-up. C2 chassis (200 V-class DC-link) driving a TECO MAX-IE3 10 HP induction motor; bus stepped 140 / 150 / 170 / 180 V.

When the bench supply hit its **current limit at ~160 s** and the bus collapsed to ~30 V, the controller rode through the dip and **resumed stable operation** — no reset, no fault. Peak ~38 A; max bus 182.5 V.

TEST CAMPAIGN	SETUP	KEY RESULT
180 V power-stage bring-up	TECO 10 HP induction motor; bus stepped 140 to 180 V	200 V-class validated; ~38 A peak; clean fault recovery
1-hour no-load endurance	C2 + induction motor, 120 VDC, ~60 Hz	60 min continuous, zero faults; thermal baseline
180 V reversal stress, 20 min	±60 Hz every ~45 s; passive cooling	26 reversals, zero faults; max ~41 A
Self-commissioning accuracy	Inverter estimates vs. LCR meter; PMSM + induction	Resistance within ~6.5 to +7.3 %; inductance +2.4 %

Six further instrumented calibration reports underpin these results — 10 in all; each links its interactive telemetry log (JSONL). All reports are currently draft and revised in the open.

■ SAFETY ENGINEERING

Six independent paths to the safe state

MAIN MCU · STM32H723ZG

- FOC motor control & application logic
- Sensor plausibility checks, fault handling
- Owns gate-drive supply kill (channel A)

SAFETY COPROCESSOR ·

STM32G474RCTx

- Independent ADC monitoring of all safety-critical signals
- CAN bus snooping · challenge/response watchdog
- Owns a fully independent kill path — no trust in the main MCU

SAFE STATE

SSO

Six-switch-open, immediate and hardware-enforced — no software ramp-down

TIM1 hardware break

<100 ns

1002 gate-drive power kill

<10 μs

Shared driver RESET

<1 μs

Passive 3.3 V rail-loss pull-down

no software

Watchdog-forced main-MCU reset

<100 ms

Independent coprocessor kill

<10 μs

ASIL D design target on the top safety goals via ASIL B(D) + ASIL B(D) decomposition across the two MCUs — ISO 26262 methodology applied as a design input, not a certification claim. Functional-safety roadmap: fault-injection, thermal, and vibration campaigns.

Hazards

17 identified at the torque / power boundary

HARA

Core platform + application profiles; TARA for cybersecurity; FMEA

Safety requirements

15 safety goals · 22 functional safety requirements

Test plans

92-test fault-injection plan; thermal & vibration campaigns planned

Standards

ISO 26262 & IEC 61800-5-2 safe-function mapping

Safety engineered from the silicon to the sheet metal.



Scan for the interactive telemetry log — every report links its data.

A university lab with a megawatt roadmap

OpenVVF is built in the Smart Power Lab at UC Santa Cruz. The lab's mission is open power electronics: hardware, firmware, and safety engineering that researchers, educators, and small manufacturers can actually study, build, and extend — the groups that sealed proprietary supply chains lock out.

Everything the project knows is published as it is learned: schematics and PCB releases, control firmware, hazard and threat analyses, and the raw telemetry behind every test claim. Sponsors and partners make the bench time, fabrication runs, and test campaigns possible.

SUPPORTED BY



■ TRAJECTORY



● IN BUILD

Gen 7.1 hardware

Revision of the control-module hardware now in build.



● PLANNED

Loaded testing & dyno

Dynamometer campaign on Gen 7.1 takes the platform from no-load validation to rated power.



● IN PROGRESS

Zero DSR integration

A Size 2 chassis is being readied for our 2018 Zero DSR electric motorcycle.



● IN PROGRESS

Size 1 + Size 2 DC-DC

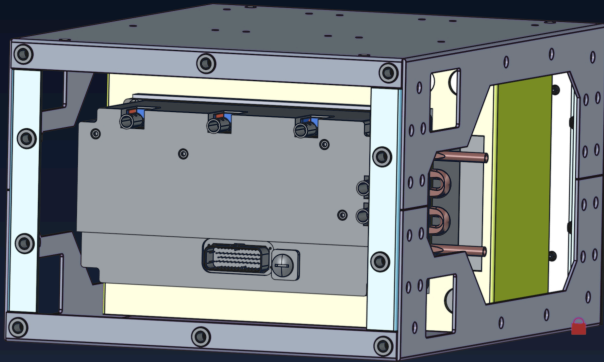
~50 kW chassis (early design) and a liquid-cooled DC-DC variant for charging, storage, and microgrids.



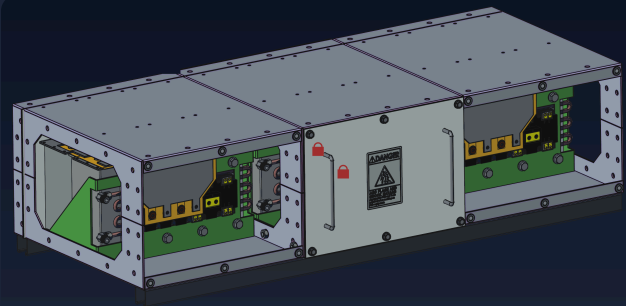
● DESIGN IN PROGRESS

Size 3 megawatt class

900 V bus, 1400 A, all-film DC-link; modular liquid-cooled enclosure with arc-flash containment analysis.



Size 2 DC-DC enclosure — design rendering. Same inverter hardware, liquid-cooled enclosure.



Size 3 megawatt-class enclosure — design rendering. Modular liquid-cooled bays, arc-flash containment analysis.

Every schematic, line of firmware, safety analysis, and test trace is public at
docs.openvvvf.org.

OPEN HARDWARE · OPEN FIRMWARE · OPEN EVIDENCE